Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/20/2006
PI Name:	Bunick, Gerard J. Ph.D.		
Project Title:	Structural Studies of Nucleosomes and Chromatin		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Oak Ridge	**State:**	TN
Zip Code:	37830-8026	**Congressional District:**	3
Comments:	Deceased September 2007.		
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	10/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	2	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG8-1826		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Detailed examination of the structure of the NCP reveals several fundamental questions regarding chromatin structure that invite further study. Microgravity-grown NCP crystals can be used to help resolve these controversies and generate further inquiries and hypotheses. To do this, the successful diffraction studies from microgravity-grown NCP crystals must be extended to provide higher resolution structures of the NCP without resorting to extraordinary methods to collect the data. Second, the experience and practical knowledge that led to the solution of the NCP structure can be used to elucidate the structures of macromolecular complexes involving the NCP in regulation of gene expression and to determine the structure of higher order chromatin elements such as transcription or repression complexes and oligonucleosome structures. In addition, this research program will pursue the production of large crystals suitable for neutron diffraction to examine the role of hydration in chromatin. The proposed research will use variants of the histone octamer and nucleosome core particle (NCP) structures to provide additional critical insights into structure-function relationships in chromatin. This is a time of great interest in the study of chromatin due to the emergence of evidence		

	linking chromatin structure as an essential factor in the regulation of chromatin function. Structures of modified nucleosome core particles are important to the elucidation of the molecular processes of gene regulation and transcription in chromatin, the substrate for all such activity in humans. The knowledge acquired from this research may provide answers to the genesis and proliferation of cancer and other diseases.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research will extend our knowledge of chromatin structural biology, providing information linking the structure of chromatin to the regulation of gene expression and control. The results may help to understand aspects of cancer and other diseases.
Task Progress:	A book “Chromatin Structure and Dynamics: State-of-the-Art” edited by Jordanka Zlatanova and Sanford Leuba has been published with a chapter titled “The Core Particle of the Nucleosome”. The 2.5 Å X-ray crystal structure of the nucleosome core particle provides significant additions to the understanding of the nucleosome, and addresses our current understanding of chromatin structure, dynamics, and function. To explore specialized types of the NCP, two H3 like centromeric histones, DH6 and YMH3 have been expressed and purified. The DH6 has been substituted for histone H3 in an octamer reconstitution so that this modified octamer can be used for nucleosome reconstitution experiments. We have also made clones for the chicken CenpA and human CenpA H3-like histone proteins. The full- length clones did not express. We modified the clones to remove a run of prolines in the N-terminal tail of both clones. The modified chicken and human CenpA clones express well. The CenpA H3-like histone will be incorporated into nucleosomes to create “centromeric” nucleosome core particles for crystallization and structure determination. These structures should be informative due to the current interest in CenpA and other H3 like histones and their role during the cell cycle. We made a clone for the non-histones high mobility group protein HMGN1 and demonstrated expression of the gene product. The HMGN1 will be complexed with nucleosome core particles. Efforts will be made to crystallize the complex for structure determination. We are refining the neutron structure of D-xylose isomerase and have completed refinement of a 0.94 Angstroms resolution X-ray structure of the enzyme. The neutron studies were the first to use spallation-produced neutrons and time-of-flight diffraction measurement on such a large macromolecule. The research has been published. Neutron and X-ray structures of the enzyme with various ligands will be determined in the future. Beam time at the Los Alamos Neutron Science Center (LANSCE) Protein Crystallography Station (PCS) is scheduled in 2005.
Bibliography Type:	Description: (Last Updated: 11/16/2007)
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Articles in Peer-reviewed Journals	Hanson, B. L.; Langan, P.; Katz, A.; Li, X.; Harp, J. M.; Glusker, J. P.; Schoenborn, B. P.; and Bunick, G. J. "Preliminary Time of Flight Neutron Diffraction Study of S. rubiginosus D-Xylose Isomerase" Acta Cryst. D60,241-249 , Feb-2004
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Books/Book Chapters	Harp, J. M.; Hanson, B. L.; and Bunick, G. J. "The Core Particle of the Nucleosome" in "Chromatin Structure and Dynamics: State-of-the-Art" Ed. Zlatanova, J; Leuba, S. H. Elsevier, 13-44 , Mar-2004
Books/Book Chapters	Hanson, B. L.; Harp, J. M.; and Bunick, G. J. "Preparation and Crystallization of the Nucleosome Core Particle" in "Chromatin and Chromatin Remodeling Enzymes - Methods in Enzymology " Ed. Allis, D.C.; Wu, C. Elsevier Academic Press, 44-62 , Jun-2004